Final Project - Depth Anything V2:
Non-training Reproduction and Analysis of Monocular Depth Estimation

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https://github.com/Oakley1001/Computer_Vision/tree/main/CV2026_Final-Project

1. Problem Definition

Monocular Depth Estimation (MDE) aims to predict a dense depth map from a single RGB image, estimating the relative depth relationship of image regions or pixels. Unlike stereo vision, Structure from Motion, or multi-view reconstruction, MDE does not require multiple images, camera intrinsic or extrinsic parameters, feature matching, or triangulation at inference time. Instead, it infers depth from semantic cues, perspective, occlusion, texture, and object shape within a single image. Therefore, MDE plays an important role in applications such as 3D reconstruction, navigation, autonomous driving, AR/VR, and AI-generated images, videos, and 3D scenes.

However, a single image lacks explicit geometric constraints, making depth prediction prone to errors in complex scenes. The paper specifically points out that existing methods still have limitations in complex layouts, transparent objects, reflective surfaces, thin objects, small holes, and object boundaries. For example, Depth Anything V1 has strong robustness, efficiency, and transferability, but it is weaker in fine details, transparent objects, and reflections. In contrast, diffusion-based methods such as Marigold can generate more detailed depth maps, but they are slower during inference and require larger models, making them less suitable for practical deployment.

The problem addressed by Depth Anything V2 [1] is how to build a monocular depth estimation foundation model that simultaneously achieves fine-grained details, robust predictions, efficient inference, and strong transferability. The core argument of the paper is that depth estimation quality depends not only on model architecture, but also heavily on the quality of training data. Real labeled depth datasets often contain label noise. For instance, depth sensors struggle with transparent objects, stereo matching is vulnerable to repetitive patterns, and SfM can be affected by dynamic objects or outliers. In addition, real depth maps often lack fine details, causing models to produce over-smoothed predictions.

Therefore, Depth Anything V2 [1] reformulates the problem as a combination of data design and model generalization. It first trains a large DINOv2-G teacher model using precise synthetic images. The teacher model then generates pseudo depth labels for 62M unlabeled real images, and student models of different scales are trained using these pseudo-labeled images. Through this pipeline, the model aims to combine the precise depth supervision of synthetic data with the scene diversity of real images, enabling monocular depth estimation to work more reliably on open-world images.

2. Motivation

Depth Anything V2 [1] is motivated by the trade-off among detail quality, robustness, and inference efficiency in existing monocular depth estimation methods. Depth Anything V1 is a discriminative model with faster inference speed, fewer parameters, and strong generalization ability. However, its depth predictions are still not detailed enough around thin structures, transparent objects, reflective surfaces, and object boundaries. In contrast, Stable Diffusion-based methods such as Marigold can generate more detailed depth maps, but their models are larger and slower, making them less suitable for real-time or resource-limited applications. Therefore, Depth Anything V2 aims to build a monocular depth estimation foundation model that achieves fine-grained details, robust predictions, and transferability at the same time.

Beyond the trade-off in model capability, the paper emphasizes the problem of training data itself. Many previous methods tend to use many labeled real images. For example, Depth Anything V1, Metric3D, and ZeroDepth all collect millions to tens of millions of labeled images. However, real depth labels often contain label noise. Labels generated by depth sensors, stereo matching, or SfM may produce unreliable annotations for transparent objects, reflective surfaces, repetitive patterns, dynamic objects, or outliers. In addition, real depth maps often lack details. Regions such as object boundaries, small holes, tree leaves, and chair legs may be over-smoothed, which limits the model's ability to generate fine-grained depth predictions.

To address this issue, Depth Anything V2 replaces labeled real images with precise synthetic images when training the large teacher model. Since synthetic images are generated by graphics engines, they can provide cleaner and more complete geometric supervision, especially around thin structures, object boundaries, transparent objects, and reflective surfaces, where real sensors or stereo-based methods may produce unreliable annotations. However, synthetic images also suffer from distribution shift and limited scene coverage, so they cannot directly guarantee generalization to real-world images.

Therefore, Depth Anything V2 further uses the large teacher model to generate pseudo depth labels for 62M unlabeled real images, and then trains student models of different scales with these pseudo-labeled real images. This design combines the precise annotations of synthetic data with the scene diversity of real images, allowing the model to produce more robust and detailed depth maps for open-world images. In addition, the authors point out that existing benchmarks suffer from label noise and limited scene diversity. To address this, they construct the DA-2K benchmark, which covers scenarios such as indoor, outdoor, transparent/reflective, aerial, and underwater scenes, providing a more comprehensive evaluation of model capability.

3. Related Work

Early monocular depth estimation methods mainly focused on metric depth estimation within specific datasets, such as the indoor NYU-D [2] dataset and the autonomous driving KITTI dataset [3]. These methods can produce reliable depth predictions in fixed domains, but their generalization ability is limited when test images come from different environments or open-world images. Therefore, later research gradually shifted toward zero-shot relative depth estimation, aiming to predict reasonable relative depth even in unseen scenes.

MiDaS [4] is a representative method in zero-shot monocular depth estimation. It trains on a mixture of multiple depth datasets, allowing the model to learn relative depth relationships across datasets. Depth Anything V1 [5] continues this data-driven direction and further uses 62M unlabeled real images to improve robustness and open-world generalization. However, Depth Anything V2 points out that Depth Anything V1 is still limited by the quality of labeled real images, since real depth labels may contain label noise and often lack details around object boundaries, thin structures, transparent objects, and reflective surfaces.
Another related direction is diffusion-based depth estimation. Marigold [6] uses the generative ability of Stable Diffusion [7] for monocular depth estimation and can produce more detailed depth maps, especially around thin objects and object boundaries. However, these methods are usually larger and slower during inference. In contrast, Depth Anything V2 adopts a discriminative model architecture based on a DINOv2 [8] encoder and a DPT depth decoder [9], aiming to preserve fine-grained details while maintaining faster inference speed and better deployment flexibility.

Depth Anything V2 is also related to synthetic-to-real transfer and pseudo-labeling. The paper uses precise synthetic images to replace noisy labeled real images when training the large teacher model, because synthetic images can provide cleaner and more complete geometric supervision, especially around thin structures, object boundaries, transparent objects, and reflective surfaces, where real sensors or stereo-based labels often fail. However, synthetic images suffer from distribution shift and limited scene coverage, so they cannot directly solve real-image generalization. To address this, Depth Anything V2 uses the large teacher model to generate pseudo depth labels for 62M unlabeled real images, and then trains smaller student models using these pseudo-labeled real images. This teacher-student training strategy is similar to knowledge distillation [10], but it is mainly performed at the label level, allowing smaller models to learn the depth prediction ability of the large model while preserving high inference efficiency.

4. Method Overview

Depth Anything V2 follows a discriminative monocular depth estimation framework based on DINOv2 encoders and a DPT depth decoder. Instead of designing a new network architecture, the paper focuses on improving the training data pipeline. The overall method consists of three stages. First, a large DINOv2-G teacher model is trained using precise synthetic images. These synthetic images provide accurate depth supervision for object boundaries, thin structures, transparent objects, and reflective surfaces. Second, the trained teacher model generates pseudo depth labels for 62M unlabeled real images. Third, student models of different scales are trained on these pseudo-labeled real images to improve robustness and generalization in open-world scenes.

The training strategy is designed to combine the advantages of synthetic and real images. Synthetic images provide clean and complete depth labels, but their distribution and scene coverage are limited. Unlabeled real images provide diverse real-world scenes, but they do not have ground-truth depth labels. By using the teacher model to annotate real images, Depth Anything V2 transfers the precise geometric knowledge learned from synthetic data to real-image distributions. This pseudo-labeling process allows smaller student models to learn high-quality depth prediction while maintaining efficient inference.

For optimization, the model uses two main loss terms on labeled synthetic images: a scale- and shift-invariant loss and a gradient matching loss. The scale- and shift-invariant loss supervises relative depth while reducing the effect of global scale and shift ambiguity. The gradient matching loss encourages sharper depth boundaries and improves fine-grained details. On pseudo-labeled real images, the method further follows Depth Anything V1 by adding a feature alignment loss to preserve semantic information from the pre-trained DINOv2 encoder. During training on pseudo-labeled images, the regions with the top 10% largest losses are ignored because they may contain noisy pseudo labels.

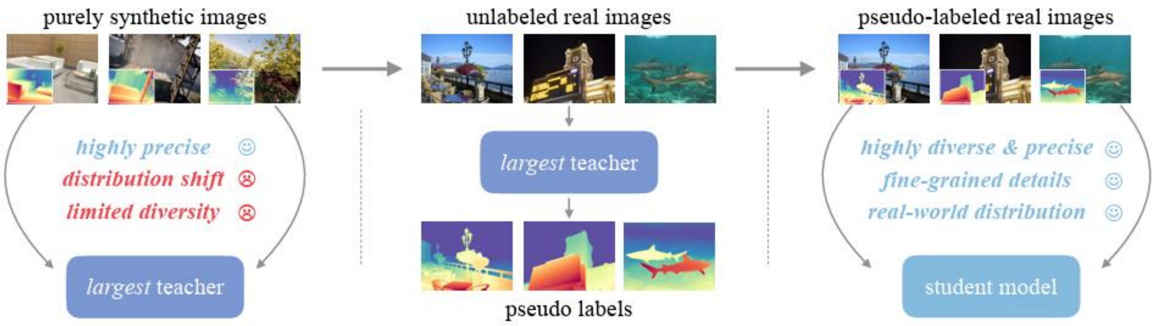

Figure 1. Training pipeline of Depth Anything V2. The model first trains a large DINOv2-G teacher on precise synthetic images. The teacher then generates pseudo-depth labels for 62M unlabeled real images. Finally, student models are trained on these pseudo-labeled real images. This design uses synthetic data for accurate depth supervision and real images for broader scene coverage, reducing the synthetic-to-real domain gap while maintaining efficient inference.

5. Experiments and Analysis

This project conducts a non-training reproduction and behavioral analysis of Depth Anything V2. We do not retrain the model or claim to reproduce the full training pipeline of the paper. Instead, we use the released pretrained checkpoints and official code to evaluate the model under several controlled inference settings. The experiments are designed to analyze the model from both quantitative and qualitative perspectives, including benchmark reproduction, model-scale comparison, resolution scaling, metric-depth inference, and point-cloud visualization.

From the camera model perspective, image formation is a projection from the 3D world to a 2D image. During this 3D-to-2D projection, geometric information such as distance and depth is lost. Depth Anything V2 addresses this missing dimension from a learning-based perspective by predicting a dense depth value for each pixel from a single RGB image. Therefore, our experiments focus not only on whether the model works, but also on what kinds of depth cues it appears to use and where its limitations remain.

In addition to the official example images, we further collect self-captured mobile-phone photos as a real-world qualitative stress test. These images include close-up objects with defocused backgrounds, long-range night scenes, strong sunlight and lens flare, low-light outdoor scenes, reflective surfaces, animals, and urban street scenes. This allows us to observe how Depth Anything V2 behaves under more realistic and uncontrolled imaging conditions.

5.1 Experimental Results

5.1.1 DA-2K Quantitative Reproduction

We first evaluate the official relative-depth checkpoints of Depth Anything V2 on the DA-2K benchmark. DA-2K contains 1033 annotated images and 2068 relative-depth point pairs. We test three model scales, ViT-S, ViT-B, and ViT-L, using the same input size of 518. Since the model predicts affine-invariant inverse depth, the evaluation checks whether the predicted inverse depth of the closer point is larger than that of the farther point.

The reproduced results are highly consistent with the reported paper numbers. ViT-S achieves 95.21% locally compared with 95.30% in the paper; ViT-B achieves 97.05% compared with 97.00%; and ViT-L achieves 97.10%, exactly matching the reported value. The differences are all around or below 0.1 percentage points, which suggests that our local setup, official checkpoints, and evaluation pipeline are reliable.

Model	Params	Paper Acc.	Local Acc.	Difference
ViT-S	24.8M	95.30%	95.21%	-0.09 pp
ViT-B	97.5M	97.00%	97.05%	+0.05 pp
ViT-L	335.3M	97.10%	97.10%	0.00 pp

Table 1. DA-2K reproduction results. Our local results closely match the reported paper numbers across all three model scales

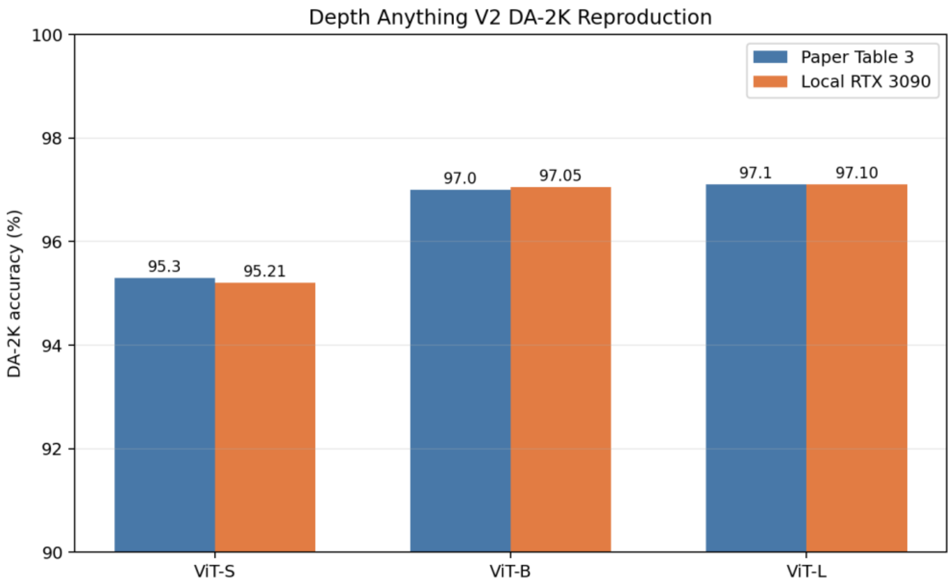

Figure 2. Comparison between paper-reported and locally reproduced DA-2K accuracy. The reproduced results are nearly identical to the reported values, validating our local evaluation pipeline.

5.1.2 Per-scenario DA-2K Analysis

In addition to the overall accuracy, we also observe the per-scenario results, including indoor, outdoor, non-real, transparent or reflective, adverse-condition, aerial, underwater, and object-centric images. The model performs strongly across most scenarios, especially in non-real, aerial, underwater, and object-centric cases. This performance is consistent with the model using semantic and structural priors in addition to local texture cues.

Indoor and outdoor scenes show relatively lower accuracy. Transparent and reflective surfaces remain physically ambiguous, although the benchmark accuracy is still high. This is reasonable because a single image cannot fully resolve real-world scale, reflection, or transparency. These cases reveal the fundamental limitation of monocular depth estimation: without multi-view geometry or direct sensor measurement, some depth relationships must be inferred rather than geometrically determined.

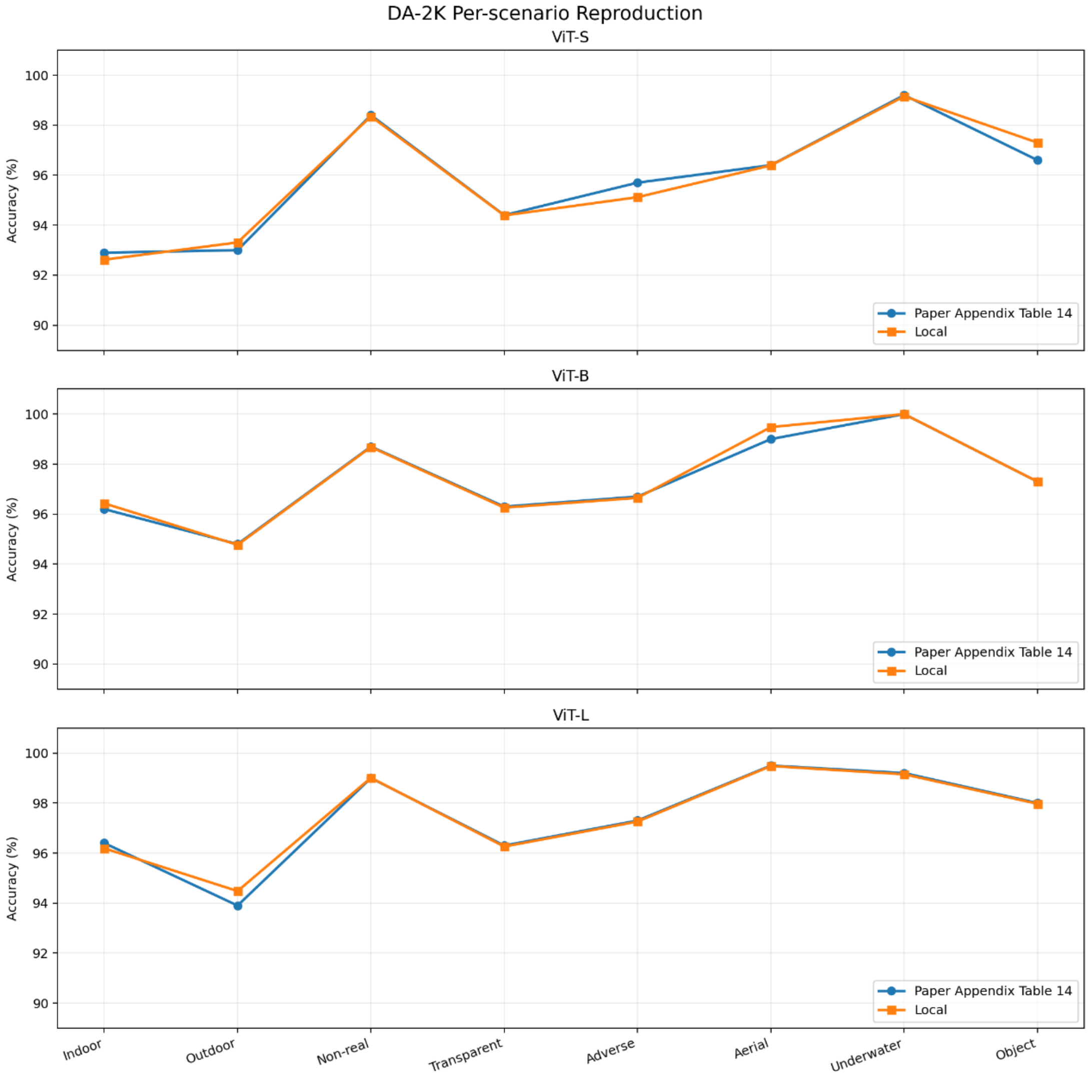

Figure 3. DA-2K per-scenario reproduction results. The local results are close to the paper-reported values across indoor, outdoor, non-real, transparent, adverse, aerial, underwater, and object-centric scenarios. This supports the robustness of the reproduced evaluation and shows that Depth Anything V2 performs consistently across diverse scene types.

5.1.3 Qualitative Relative-depth Results

We then perform qualitative inference on 20 official example images using ViT-S, ViT-B, and ViT-L at input size 518. The examples include outdoor street scenes, indoor rooms, paintings, line drawings, transparent objects, object-centric images, and complex structures. The goal is to observe whether the model produces reasonable foreground-background ordering, object boundaries, and global scene layout.

The visual results show that Depth Anything V2 usually predicts coherent relative-depth maps.

In street scenes, nearby vehicles, pedestrians, and road regions are predicted as closer, while the sky, distant buildings, and background regions are predicted as farther. In indoor scenes, the model can generally separate tables, chairs, walls, doors, windows, and floors into reasonable depth layers.

A particularly interesting observation is that the model also predicts plausible depth maps for paintings and line drawings. These inputs do not contain physically accurate texture, lighting, or shading, so the results indicate that Depth Anything V2 strongly relies on learned semantic and structural priors. This is a strength because it allows the model to generalize beyond ordinary natural images. At the same time, it is also a limitation: the output may look visually plausible even when it is not physically or metrically correct.

5.1.4 Self-captured Image Stress Test

In addition to the official example images, we further test Depth Anything V2 on mobile-phone images. Unlike the official examples, these images are taken under more realistic and uncontrolled imaging conditions. The selected cases include close-up objects with defocused backgrounds, urban street scenes, low-light night scenes, strong sunlight and lens flare, reflective or transparent surfaces, through-window views, animals, vehicles, large-scale outdoor objects, and distant landscapes. Since these images do not have ground-truth depth annotations, this experiment is used as a qualitative stress test rather than a quantitative evaluation.

Close-up objects with defocused backgrounds

The first group contains close-up scenes such as animals, food, and small objects photographed at short distance. These images test whether the model can separate the foreground object from a defocused background. In most cases, the model predicts the main object as closer and places the blurred background farther away in the relative-depth ordering. This suggests that Depth Anything V2 can use object size, occlusion, focus blur, and semantic object cues to infer foreground-background relationships.

However, the results also show that defocus blur may reduce local depth reliability. When the background is heavily blurred, the model can still produce a plausible global layout, but small structures in the blurred regions become less detailed.

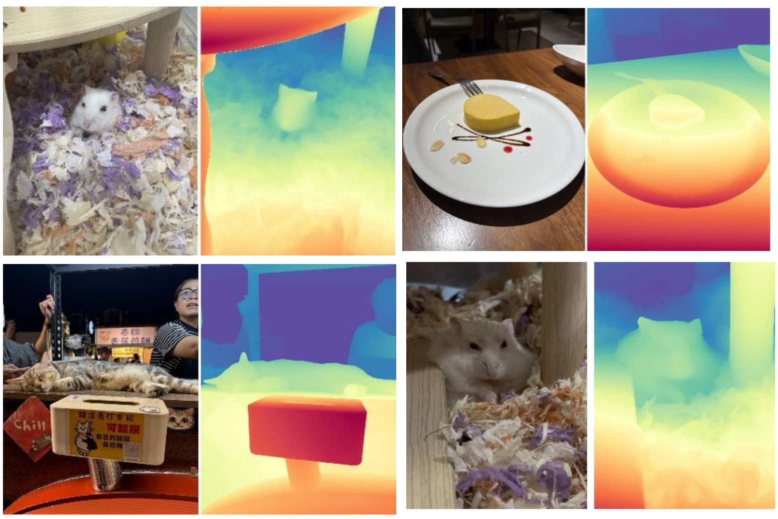

Figure 4(a). Close-up objects with defocused backgrounds. The examples include animals, food, and nearby objects. The model usually separates the foreground object from the background, while blurred regions contain weaker local depth details.

Urban street and object-scale scenes

The second group contains urban street scenes, buildings, cars, alleys, and pedestrian-scale environments. These images test whether the model can infer global scene layout from perspective, object scale, road structure, and occlusion. The results show that nearby roads, cars, street objects, and foreground structures are generally predicted as closer, while distant buildings and sky regions are predicted as farther.

This behavior indicates that Depth Anything V2 can capture common geometric cues in urban scenes, such as vanishing direction, building layout, and foreground-background ordering. However, local boundaries around thin objects, trees, poles, and vehicles may still be over-smoothed, especially when the scene contains strong shadows or complex occlusion.

Figure 4(b). Urban Street and Object-scale Scenes. The model captures the global depth layout of roads, buildings, cars, and alleys. Thin structures and cluttered regions remain more challenging.

Low-light and long-range night scenes

The third group contains low-light night scenes and long-range city views. These cases are more difficult because distant buildings, city lights, and the night sky provide weak geometric cues. In these images, the model still predicts a general near-to-far structure, but far-away regions are often compressed into a similar depth range.

This limitation is reasonable from a monocular depth estimation perspective. When the scene is very far away, perspective, texture, and familiar-scale cues become weak, while low-light conditions further reduce texture and boundary clarity. As a result, the predicted depth map becomes less reliable for small buildings, city lights, and horizon regions.

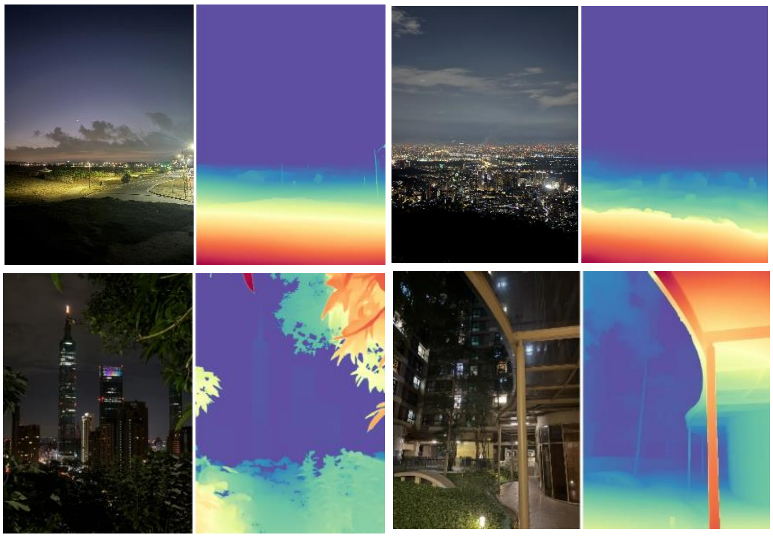

Figure 4(c). Low-light and long-range night scenes. The model can produce a plausible global depth gradient, but distant buildings, city lights, and the night sky are often compressed into similar depth values.

Low-light large-scale outdoor scenes

The fourth group contains low-light large-scale outdoor scenes, such as night-time cruise decks and dark coastal scenes. Compared with ordinary night city views, these images contain large open areas and weak texture regions. The model can still infer a broad foreground-background relationship, such as nearby decks or ground regions being closer and distant structures being farther.

However, because large dark regions contain limited visual information, the predicted depth often becomes smooth. Fine details in the far background are less reliable, and the model tends to rely on large scene layout rather than precise local geometry.

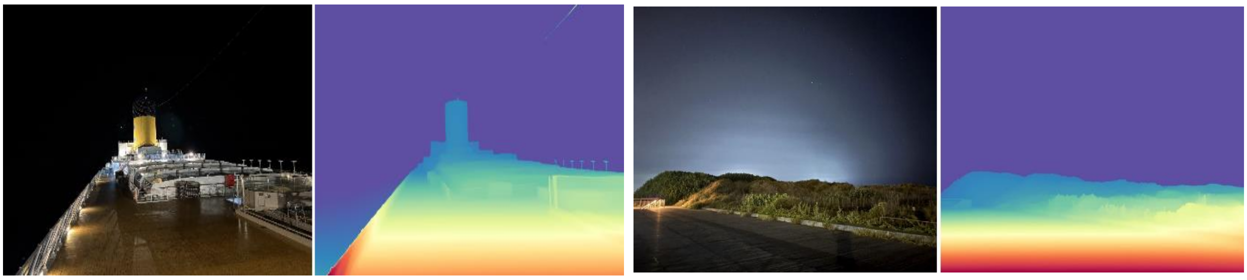

Figure 4(d). Low-light large-scale outdoor scenes. The model preserves the broad depth layout in night-time outdoor scenes, but dark and weak-texture regions reduce fine-grained depth reliability.

Strong illumination, glare, and lens flare

The fifth group contains images with strong sunlight, glare, overexposed regions, reflections on water, and lens flare. These cases test whether the model is affected by illumination artifacts. In several examples, the model still captures the main scene layout, such as nearby ground, railings, trees, water surfaces, and distant background regions.

However, glare and lens flare can introduce visual patterns that do not correspond to real 3D geometry. Light streaks, overexposed regions, or reflections may be interpreted as scene structures or may cause overly smooth depth transitions. This shows that the model is still sensitive to image appearance when illumination artifacts conflict with physical depth.

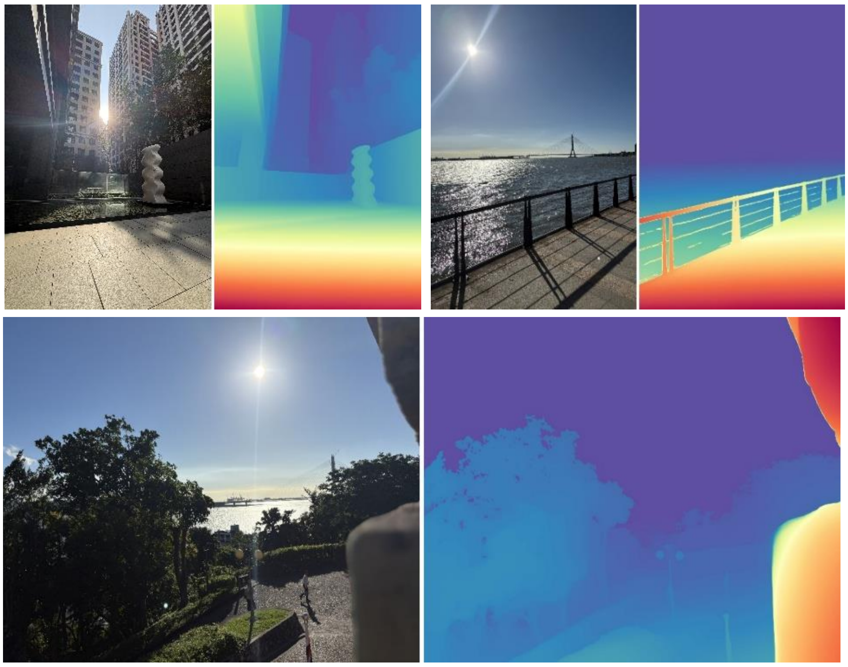

Figure 4(e). Strong Illumination, Glare, and Lens Flare. The model preserves the main foreground-background relationship, but strong sunlight, glare, reflections, and light streaks may affect local depth boundaries.

Reflective, transparent, and through-window scenes

The sixth group contains images taken through windows or involving reflective surfaces, such as window views, mirrors, glass, and airplane-window scenes. These images are challenging because reflections and transparent surfaces mix visual information from different depths. For example, a window may show both the outside scene and reflections from the inside environment.

The results show that the model can usually produce a visually plausible depth map, but the prediction may not correspond to the actual physical surface depth. This is a common limitation of monocular depth estimation because the image appearance of reflective or transparent regions does not directly reveal the true geometry.

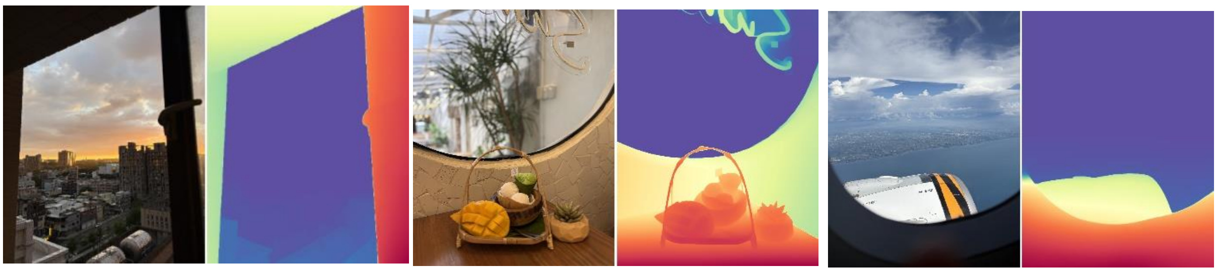

Figure 4(f). Reflective, transparent, and through-window scenes. The model produces plausible relative-depth maps, but glass, mirrors, and window reflections make the physical depth interpretation ambiguous.

We further tested the model on large-scale outdoor scenes with long-range depth variation, such as a coastal scene with a distant island and a cruise-ship harbor scene. These examples contain broad foreground-to-background transitions, including sea surfaces, human-scale structures, large objects, and distant background regions. The predicted depth maps show that Depth Anything V2 captures the global depth ordering reasonably well. In particular, the cruise-ship example demonstrates that the model can separate the large foreground structure from the background sky and sea, while the island example shows that the model preserves the near-far layering even when the distant target occupies only a small image region. However, distant regions remain relatively smooth, suggesting that the model captures coarse global structure more reliably than fine depth variation in very far areas.

Figure 4(g). Large-scale outdoor and distant scenes. The model captures broad foreground-background structure in coastal and harbor scenes, such as a distant island and a cruise ship. However, very distant sky, sea, and horizon regions remain smooth and contain limited fine-grained depth variation.

Overall, the self-captured image stress test shows that Depth Anything V2 generalizes well to real-world mobile-phone photos. It performs especially well in preserving foreground-background ordering and global scene layout. Nevertheless, the results are affected by monocular ambiguity, background defocus, low light, glare, reflections, transparent surfaces, and very large depth ranges. Therefore, these outputs should be interpreted as qualitative relative-depth estimates rather than accurate metric 3D measurements.

5.1.5 Model-scale Comparison

We compare ViT-S, ViT-B, and ViT-L on the same set of 20 images. Overall, the three models produce similar global depth layouts, meaning that even the smaller ViT-S model captures the main foreground-background relationship of the scene. However, larger models generally preserve sharper object boundaries and more stable fine structures.

The difference is most visible in difficult regions such as railings, chair legs, line drawings, thin object contours, and cluttered indoor scenes. ViT-L usually gives cleaner boundaries and finer local details, while ViT-S may produce smoother or less precise boundaries. This suggests that model scaling mainly improves local detail and boundary quality rather than completely changing the global depth interpretation.

In terms of computational cost, the 20-image inference took 8.30 seconds for ViT-S, 9.64 seconds for ViT-B, and 14.57 seconds for ViT-L in our local environment. This timing includes conda startup, checkpoint loading, image I/O, visualization, and inference, so it should not be interpreted as pure GPU latency. Nevertheless, it reflects a practical trade-off: larger models provide better detail but require more computation.

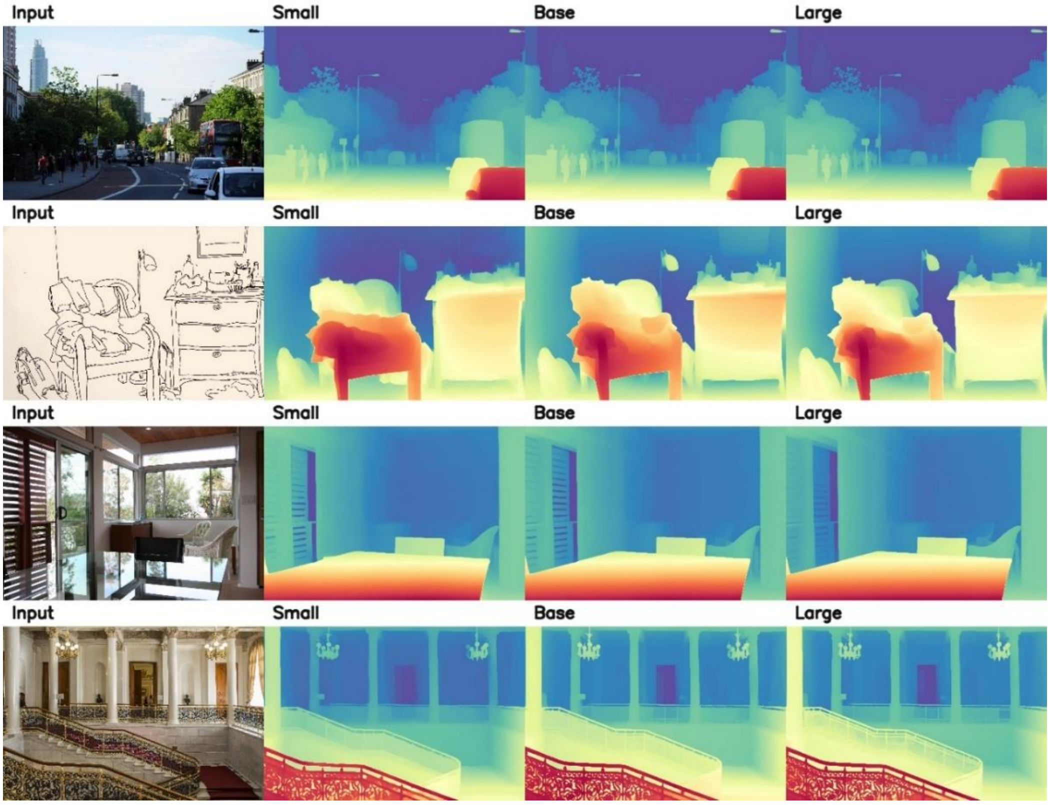

Figure 5. Model-scale comparison. ViT-S, ViT-B, and ViT-L produce similar global depth layouts, while larger models preserve sharper boundaries and finer structures.

5.1.6 Resolution-scaling Comparison

We further test the effect of input resolution by running ViT-L at input sizes 518 and 1036 on four selected images: an outdoor street scene, a line drawing, an indoor room, and an architectural indoor scene. This experiment examines whether increasing test-time resolution improves the predicted depth map.

The results show that higher input resolution mainly improves local details rather than changing the overall depth ordering. Thin structures such as railings, tree branches, line strokes, and object boundaries become clearer at input size 1036. However, the global interpretation of which regions are near or far remains mostly consistent with the 518-resolution result.

This suggests that Depth Anything V2's global depth layout is mainly determined by high-level scene understanding, while higher resolution helps recover high-frequency details. Therefore, resolution scaling is useful when boundary sharpness is important, but it does not solve fundamental monocular ambiguity.

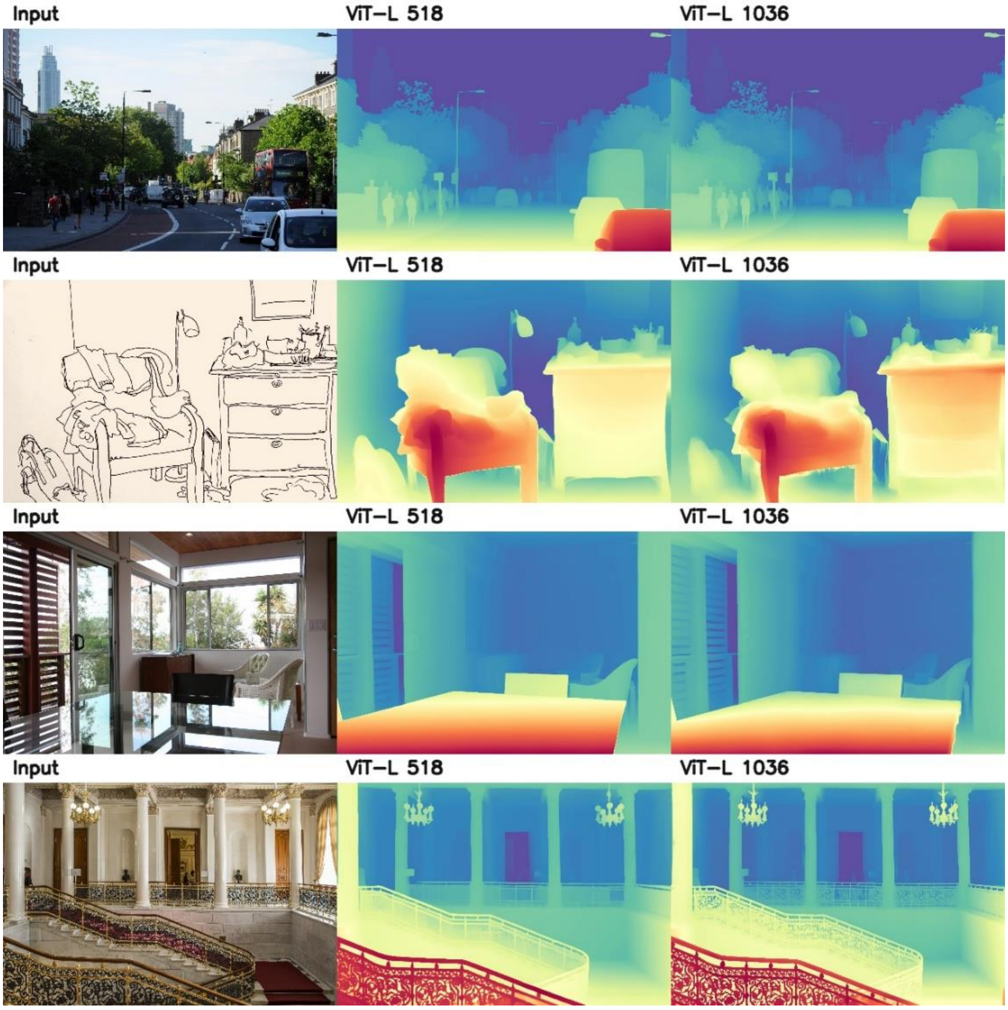

Figure 6. Resolution-scaling comparison. Higher input resolution improves local details and thin structures, but the global depth ordering remains mostly unchanged.

5.1.7 Metric-depth Inference and Point-cloud Visualization

Finally, we evaluate the metric-depth checkpoints and convert one predicted depth map into a 3D point cloud. This part connects the project to the course topic of 3D representations. A depth map is a 2.5D representation because it stores visible depth values associated with image pixels, but it does not capture the full 3D structure behind occluded regions. Point clouds, on the other hand, are unordered sets of 3D points and are commonly used as a 3D representation, although they do not contain explicit surface connectivity.

For metric-depth inference, we use the Hypersim checkpoint for the indoor image with a maximum depth of 20 m, and the Virtual KITTI 2 checkpoint for the outdoor image with a maximum depth of 80 m. The indoor result has an average predicted depth of about 3.64 m, while the outdoor result has an average predicted depth of about 34.31 m. These values describe the prediction distributions only. Since the images do not have ground truth depth, they do not measure metric accuracy. This experiment verifies that the separately fine-tuned official metric depth checkpoints can produce meter scaled predictions. However, metric depth is more domain-dependent than relative depth, since indoor and outdoor scenes require different checkpoints and depth ranges.

We then convert the indoor metric-depth prediction into a colored point cloud. The point cloud roughly preserves the visible room layout, demonstrating how monocular depth estimation can be connected to 3D scene representation. However, it remains a single-view reconstruction: hidden regions cannot be recovered, and the result depends on assumed camera intrinsics. In implementation, our attempt to generate a ViT-L point cloud caused CUDA out-of-memory on an RTX 3090 24 GB because the official script used the original image height as inference size. Therefore, the final point-cloud visualization uses the ViT-S metric-depth checkpoint.

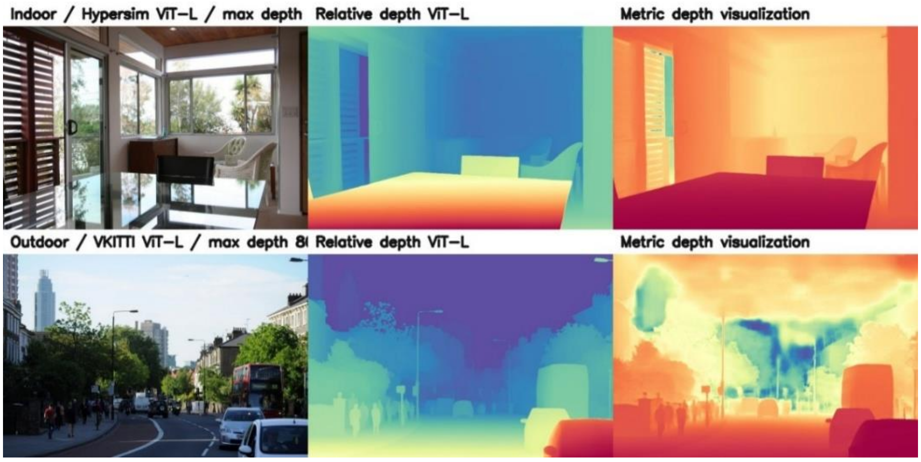

Figure 7(a): metric depth comparison

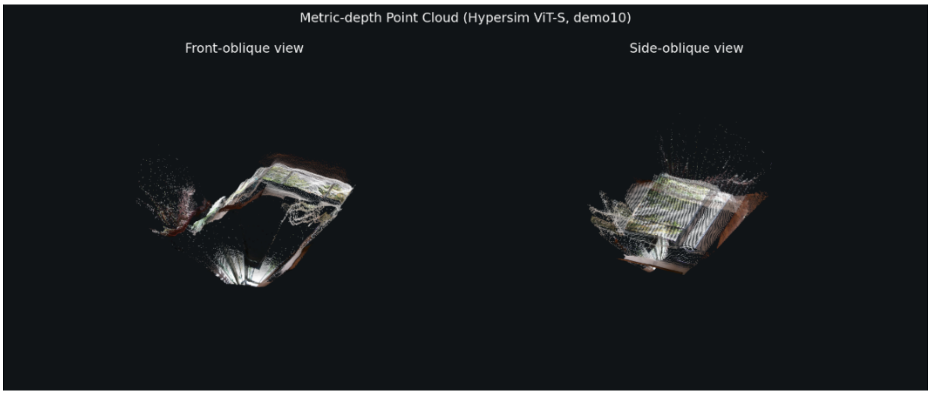

Figure 7(b): point cloud render

Figure 7. Metric-depth and point-cloud extension.

(a) Metric-depth comparison between indoor and outdoor examples. Relative depth only represents foreground-background ordering, while metric depth predicts values in meters using domain-specific checkpoints.

(b) Point-cloud visualization converted from the indoor metric-depth result. The point cloud preserves the visible room layout, but it remains a single-view 3D representation and cannot recover occluded geometry.

5.1.8 Video Inference Verification

We also tested the official video inference function using two sample videos, ferris_wheel.mp4 and basketball.mp4. The model processes each frame independently and produces relative-depth visualizations for dynamic scenes. This experiment verifies that the released implementation can be applied to video inputs, not only still images.

However, since this setting does not include a temporal consistency module, we treat the result as a functional verification rather than a formal video-depth evaluation. In other words, this experiment shows that Depth Anything V2 can generate frame-wise depth maps for videos, but it does not evaluate whether the predicted depth is temporally stable across consecutive frames.

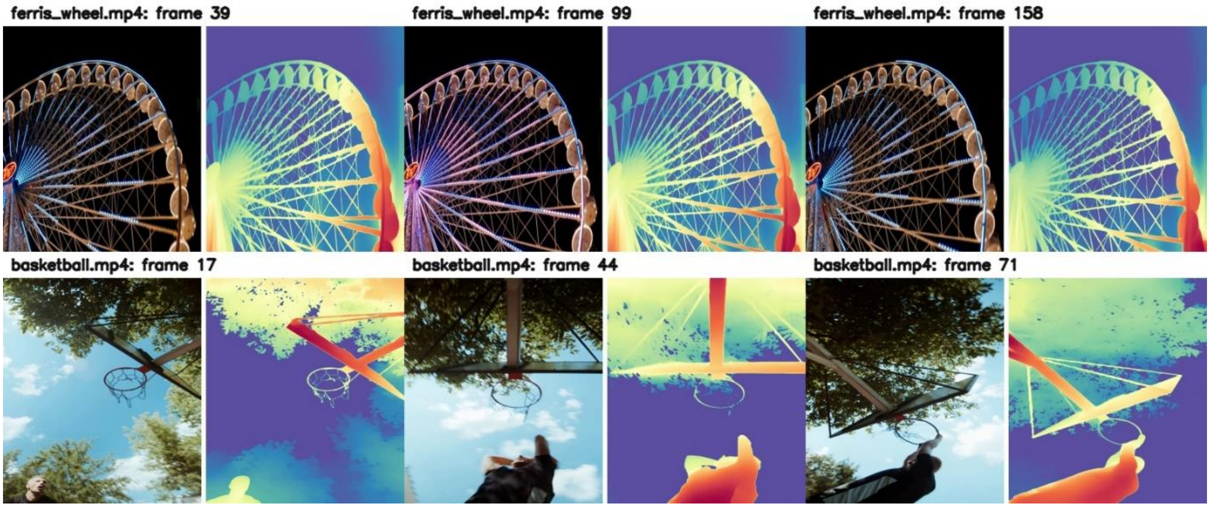

Figure 8. Video inference keyframes. The model can generate frame-wise relative-depth maps for dynamic scenes, but this experiment does not evaluate temporal consistency.

5.2 Discussion

5.2.1 Quantitative Reproduction Reliability

The DA-2K reproduction is the most important quantitative validation in this project. Since the local results are nearly identical to the paper-reported numbers, the experiment confirms that the official pretrained models and our evaluation pipeline are reliable. This confirms the correctness of our checkpoints and evaluation pipeline, providing a reliable basis for the following qualitative and behavioral analyses. Instead of only showing demo images, the benchmark reproduction provides numerical evidence that the model behaves consistently with the paper.

5.2.2 Model Behavior and Learned Depth Priors

The qualitative results show that Depth Anything V2 can produce reasonable depth maps not only for natural photographs, but also for paintings and line drawings. This suggests that the model is not simply performing low-level texture analysis. Instead, it appears to rely on learned semantic and structural priors, such as object shape, perspective layout, occlusion relationships, and typical scene structure.

This behavior explains why the model can generalize to diverse open-world images. However, it also means that the predicted depth may be visually plausible rather than physically correct. In other words, Depth Anything V2 estimates the most likely depth structure based on learned priors, but it does not recover depth through explicit geometric constraints like stereo vision or Structure from Motion.

The mobile-phone images further support this observation. In close-up scenes with background defocus, the model can usually separate the foreground object from the blurred background, suggesting that it uses both semantic object cues and photographic cues such as focus blur. In outdoor scenes with strong sunlight or lens flare, the model still captures the main foreground-background relationship, but illumination artifacts sometimes affect boundary quality and depth smoothness. In very distant night scenes, the predicted depth becomes less reliable because most distant regions contain weak geometric cues and are compressed into a similar depth range.

5.2.3 Model Scale and Resolution Trade-off

The model-scale comparison shows that ViT-S, ViT-B, and ViT-L produce similar global depth layouts. This indicates that even the smaller model is already capable of capturing the main scene structure. Larger models mainly improve object boundaries, thin structures, and fine local details. Therefore, model scale affects the quality of local refinement more than the overall depth ordering.

The resolution-scaling experiment shows a similar trend. Increasing the input size from 518 to 1036 improves high-frequency details such as railings, line strokes, tree branches, and object contours, but it does not significantly change the global depth layout. These results suggest that both model scale and input resolution improve detail quality, but neither fully resolves the inherent ambiguity of monocular depth estimation.

5.2.4 Relative Depth, Metric Depth, and 3D Representation

The metric-depth and point-cloud experiments demonstrate how monocular depth estimation can be extended toward 3D scene understanding. Relative depth only describes the ordering of near and far regions, while metric depth attempts to predict values in meters. However, metric depth is more domain-dependent because indoor and outdoor scenes require different checkpoints and depth ranges.

The point-cloud visualization further shows the connection between depth maps and 3D representations. By back-projecting the predicted metric depth into 3D space, we can obtain a visible single-view point cloud. However, this representation remains incomplete because only visible surfaces can be reconstructed. Occluded regions are missing, and the result depends on assumed camera intrinsics. Therefore, the point cloud should be interpreted as a partial 3D visualization rather than a complete geometric reconstruction.

5.2.5 Failure Cases and Limitations

Despite its strong generalization ability, Depth Anything V2 still inherits the fundamental limitations of monocular depth estimation. Transparent and reflective objects are difficult because their appearance does not directly correspond to the physical surface depth. Outdoor scenes can also be challenging because real-world scale is ambiguous from a single image. Thin structures and small holes may be improved by larger models or higher resolution, but they are still sensitive to visual details and model capacity.

Our self-captured stress test shows similar limitations: low-light night scenes, lens flare, glare, reflective surfaces, and strong background defocus can reduce the reliability of local depth details.

Compared with stereo vision or multi-view reconstruction, Depth Anything V2 does not use test-time correspondences, epipolar geometry, or triangulation. This makes the method more flexible and easier to deploy, but also means that the result is not fully constrained by geometry. Therefore, its output should be interpreted as a learned depth estimate rather than a complete 3D measurement.

For video inputs, our experiment only verifies frame-wise inference. Since no temporal consistency module is applied, possible depth flickering across frames is not formally evaluated.

5.2.6 Connection to Course Topics

This project is closely related to several topics covered in class. The camera model lecture explains that images are formed by projecting 3D scenes onto a 2D image plane, causing depth information to be lost. The two-view geometry lecture shows that stereo and multi-view methods can reduce this ambiguity using correspondences, epipolar geometry, disparity, and triangulation. In contrast, Depth Anything V2 estimates depth from a single image using learned priors.

The project is also connected to 3D representations. The predicted depth map is a 2.5D representation because it assigns a depth value to each visible pixel. The point-cloud experiment further converts metric depth into an unordered set of 3D points, showing how depth prediction can be used as a bridge between 2D images and 3D scene representation. Overall, this project demonstrates how a modern learning-based model addresses the classical computer vision problem of recovering depth from images.

5.3 Summary

Overall, our experiments show that Depth Anything V2 is strong at producing globally consistent relative-depth maps across diverse image types, including natural images, indoor scenes, outdoor scenes, paintings, sketches, object-centric images, self-captured mobile-phone photos, and video frames. The DA-2K reproduction confirms that the released pretrained models can closely reproduce the reported benchmark performance. Model scaling and resolution scaling mainly improve boundary sharpness and local details, while the global depth layout remains stable. The video inference experiment further shows that the official implementation can process dynamic scenes frame by frame, although temporal consistency is not formally evaluated.

However, the method still inherits the fundamental limitations of monocular depth estimation. Transparent and reflective objects introduce ambiguity, metric depth depends on the training domain, and single-view point-cloud conversion cannot recover hidden geometry. Therefore, Depth Anything V2 is effective as a general-purpose depth foundation model, but its predictions should be interpreted as learned estimates rather than fully geometry-constrained 3D measurements. This also reflects the core course theme that depth is lost during camera projection, and recovering it from a single image requires either geometric constraints or learned priors.

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